AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 9: Conditioning of Linear Systems; Backward Stability of Gaussian Elimination

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Outline

1. **Condition Number of Linear Systems (NLA §22 & MC §3.3)**
2. Backward Stability of LU Factorization (NLA §22 & MC §3.3)
3. Putting It All Together

Condition Number of Linear System

Theorem

Let A be nonsingular, and let x and $\hat{x} = x + \delta x$ be the solutions of $Ax = b$ and $A\hat{x} = b + \delta b$, respectively. Then

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta b\|}{\|b\|},$$

and there exist b and δb for which the equality holds.

- *Proof sketch.* $\|\delta x\| \leq \|A^{-1}\| \|\delta b\|$ and $\|b\| \leq \|A\| \|x\|$.
- **Question:** For what b and δb is the equality achieved?

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- *Proof sketch.* $\|\delta x\| \leq \|A^{-1}\|\|\delta b\|$ and $\|b\| \leq \|A\|\|x\|$.
- **Question:** For what b and δb is the equality achieved?
  Answer: When b is in the direction of *minimum* magnification of A^{-1}, and δb is in the direction of *maximum* magnification of A^{-1}.
  In the 2-norm, when b is in the direction of *maximum* magnification of $A^{\top}$, and δb is in the direction of *minimum* magnification of $A^{\top}$.
- A matrix is *nearly singular* if its condition number is very large.

Ill Conditioning Caused by Poor Scaling

- Some matrices are ill conditioned merely because they are out of scale.

Theorem

Let $A \in \mathbb{R}^{n\times n}$ be any nonsingular matrix, and let a_k, $1 \le k \le n$ denote the kth column of A. Then for any i and j with $1 \le i, j \le n$, $\kappa_p(A) \ge \|a_i\|_p / \|a_j\|_p$.

- *Proof sketch.* $\|A\|_p \ge \|a_i\|_p$ and $\|A^{-1}\|_p \ge 1/\|a_i\|_p$ for $1 \le i \le n$.
- This theorem indicates that poor scaling inevitably leads to ill conditioning.
- A *necessary* condition for a matrix to be well conditioned is that all of its rows and columns are of roughly the same magnitude.

Non-singularity of Perturbed Matrix

Theorem

If $\boldsymbol{A}$ is nonsingular and

$$\|\delta A\|/\|A\| < 1/\kappa(A),$$

then $\boldsymbol{A} + \delta \boldsymbol{A}$ is nonsingular.

Proof

$\|\delta A\|/\|A\| < 1/\kappa(A)$ is equivalent to $\|\delta A\|\|A^{-1}\| < 1$. Suppose $A + \delta A$ is singular. Then there exists $y \neq 0$ such that $(A + \delta A)y = 0$, and $y = -A^{-1}\delta A y$. Therefore, $\|y\| \leq \|A^{-1}\|\|\delta A\|\|y\|$, or $\|A^{-1}\|\|\delta A\| \geq 1$.

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- If $A + \delta A$ is the singular matrix closest to A, in the sense that $\|\delta A\|_2$ is as small as possible, then $\|\delta A\|_2/\|A\|_2 = 1/\kappa_2(A)$.

Linear System with Perturbed Matrix

- Suppose $Ax = b$ and $\hat{A}\hat{x} = b$ where $\hat{A} = A + \delta A$. Let $\delta x = \hat{x} - x$ and $\hat{x} = x + \delta x$.
- We would like to bound $\|\delta x\|/\|x\|$, but first we bound $\|\delta x\|/\|\hat{x}\|$.

Theorem

If A is nonsingular, and let $b \neq 0$, then $\|\delta x\|/\|\hat{x}\| \leq \kappa(A)\|\delta A\|/\|A\|$.

Proof

Rewrite $(A + \delta A)\hat{x} = b$ as $Ax + A\delta x + \delta A\hat{x} = b$, where $Ax = b$. Therefore,

$$\|\delta x\| \leq \|A^{-1}\|\|\delta A\|\|\hat{x}\|.$$

Therefore,

$$\frac{\|\delta x\|}{\|\hat{x}\|} \leq \|A^{-1}\|\|\delta A\| = \kappa(A)\frac{\|\delta A\|}{\|A\|}.$$

Linear System with Perturbed Matrix Cont'd

- Suppose $Ax = b$ and $\hat{A}\hat{x} = b$, where $\hat{A} = A + \delta A$. Let $\delta x = \hat{x} - x$ and $\hat{x} = x + \delta x$.

Theorem

If A is nonsingular and $\|\delta A\|/\|A\| < 1/\kappa(A)$, and let $b \neq 0$, then

$$\frac{\|\delta x\|}{\|x\|} \leq \frac{\kappa(A)\|\delta A\|/\|A\|}{1 - \kappa(A)\|\delta A\|/\|A\|}.$$

Proof

$\|\delta x\| \leq \|A^{-1}\|\|\delta A\|\|\hat{x}\| \leq \|A^{-1}\|\|\delta A\|(\|x\| + \|\delta x\|)$. Therefore,

$$\left(1 - \|A^{-1}\|\|\delta A\|\right)\|\delta x\| \leq \|A^{-1}\|\|\delta A\|\|x\|,$$

where $\|A^{-1}\|\|\delta A\| = \kappa(A)\|\delta A\|/\|A\|$.

We typically expect $\kappa(A)\|\delta A\| \ll \|A\|$, so the denominator is close to 1.

Perturbed RHS and Matrix

- Suppose $Ax = b$ and $\hat{A}\hat{x} = \hat{b}$, where $\hat{A} = A + \delta A$, $\hat{b} = b + \delta b$, and $\hat{x} = x + \delta x$.

Theorem (Thm 2.3.8, Fundamentals of Matrix Computations, 3rd ed., David S. Watkins)

Suppose A is nonsingular, $\hat{x} \neq 0$, and $\hat{b} \neq 0$. Then

$$\frac{\|\delta x\|}{\|\hat{x}\|} \leq \kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|\hat{b}\|} + \frac{\|\delta A\|}{\|A\|}\frac{\|\delta b\|}{\|\hat{b}\|}\right).$$

Perturbed RHS and Matrix Cont'd

Proof

$A\delta x = \delta b - \delta A\hat{x}$. Hence $\|\delta x\| \le \|A^{-1}\|(\|\delta A\|\|\hat{x}\| + \|\delta b\|)$, and

$$\frac{\|\delta x\|}{\|\hat{x}\|} \le \kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|\hat{x}\|\|A\|}\right).$$

Furthermore,

$$\frac{1}{\|\hat{x}\|\|A\|} \le \frac{\|\hat{A}\|}{\|A\|\|\hat{b}\|} \le \frac{\|A\| + \|\delta A\|}{\|A\|\|\hat{b}\|} = \frac{1}{\|\hat{b}\|} + \frac{\|\delta A\|}{\|A\|\|\hat{b}\|}.$$

Perturbed RHS and Matrix

We can simplify the previous theorem to be

$$\frac{\|\delta x\|}{\|\hat{x}\|} \lesssim \kappa(A)\left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|\hat{b}\|}\right),$$

and then obtain the following result.

Theorem

If A is nonsingular and $\|\delta A\|/\|A\| < 1/\kappa(A)$, and let $b \neq 0$, then

$$\frac{\|\delta x\|}{\|x\|} \lesssim \frac{\kappa(A)(\|\delta A\|/\|A\| + \|\delta b\|/\|b\|)}{1 - \kappa(A)\|\delta A\|/\|A\|}.$$

Roughly speaking, $\kappa(A)$ determines the extra loss of digits of accuracy in x in addition to input errors in A and b.

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Stability of LU without Pivoting

- For $A = LU$ computed without pivoting (Thm 22.1 of NLA),

$$\tilde{L}\tilde{U} = A + \delta A, \qquad \frac{\|\delta A\|}{\|L\|\|U\|} = \mathcal{O}(\epsilon_{\text{machine}})$$

  (Thm 3.3.1 of *MC*, 4th ed., Golub & Van Loan).
- This is close to backward stability, except that we have $\|L\|\|U\|$ instead of $\|A\|$ in the denominator.
- Instability of Gaussian elimination can happen only if one or both of the factors L and U is large relative to the size of A.
- Unfortunately, $\|L\|$ and $\|U\|$ can be arbitrarily large (even for well-conditioned A), e.g.,

$$A = \begin{bmatrix} 10^{-20} & 1 \\ 1 & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 10^{20} & 1 \end{bmatrix} \begin{bmatrix} 10^{-20} & 1 \\ 0 & a_{22} - 10^{20} \end{bmatrix}.$$

- If $\text{fl}(a_{22} - 10^{20}) = -10^{20}$, we obtain the same L and U regardless of a_{22}.
- L and U are not exact for a nearby A, so the algorithm is *unstable*.

Stability of LU with Partial Pivoting

- With pivoting, all entries of L are in $[-1, 1]$, so $\|L\| = \mathcal{O}(1)$.
- To measure growth in U, we introduce the growth factor $\rho = \frac{\max_{i,j} |u_{ij}|}{\max_{i,j} |a_{ij}|}$, and hence $\|U\| = \mathcal{O}(\rho \|A\|)$.
- We then have $PA = LU$ and

$$\tilde{L}\tilde{U} = \tilde{P}A + \delta A, \qquad \frac{\|\delta A\|}{\|A\|} = \mathcal{O}(\rho \epsilon_{\text{machine}}).$$

- If $|\ell_{ij}| < 1$ for each $i > j$ (i.e., there is no tie for the pivoting), then $\tilde{P} = P$ for sufficiently small $\epsilon_{\text{machine}}$.
- If $\rho = \mathcal{O}(1)$, then the algorithm is backward stable.
- In fact, $\rho \leq 2^{n-1}$, so by definition ρ is a constant but can be very large.

The Growth Factor

- ρ can indeed be as large as 2^{n-1}. Consider the matrix

$$\begin{bmatrix} 1 & & & & 1 \\ -1 & 1 & & & 1 \\ -1 & -1 & 1 & & 1 \\ -1 & -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & & 0 \\ -1 & 1 & & & 0 \\ -1 & -1 & 1 & & 0 \\ -1 & -1 & -1 & 1 & 0 \\ -1 & -1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & & 1 \\ & 1 & & & 2 \\ & & 1 & & 4 \\ & & & 1 & 8 \\ & & & & 16 \end{bmatrix},$$

  where the growth factor is $\rho = 16 = 2^{n-1}$.
- $\rho = 2^{n-1}$ is as large as ρ can get. It can be catastrophic in practice.
- Theoretically, Gaussian elimination with partial pivoting is backward stable according to the formal definition.
- However, in the worst case, Gaussian elimination with partial pivoting may be unstable for practical values of n.

The Growth Factor in Practice

- **Good news**: Large ρ occurs only for very skewed matrices. Experimentally, one rarely sees a very large ρ.
- The probability of a large ρ decreases exponentially in ρ.
- "If you pick a billion matrices at random, you will almost certainly not find one for which Gaussian elimination is unstable."
- In practice, ρ is no larger than $\mathcal{O}(\sqrt{n})$. However, this behavior is not fully understood yet.
- In conclusion:
  - Gaussian elimination with partial pivoting is backward stable.
  - In theory, its error may grow exponentially in n.
  - In practice, it is stable for matrices of practical interest.

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Accuracy of Linear Solver

- Solving $Ax = b$ using LU factorization with partial pivoting is also backward stable.
  1. $PA = LU$.
  2. $Ly = Pb$.
  3. $Ux = y$.
- Each step is backward stable (we omit the detailed proof).
- The overall growth factor of the error is bounded by the product of the growth factors of the individual steps.

A Posteriori Error Analysis Using Residual

- Suppose $\hat{x}$ is a computed solution of $Ax = b$, and the residual is $\hat{r} = b - A\hat{x}$.
- Let A be nonsingular and $b \neq 0$. Then $\frac{\|\delta x\|}{\|x\|} \leq \mathcal{O}(\kappa(A))\frac{\|\hat{r}\|}{\|b\|}$.
- If the residual is tiny and A is well conditioned, then $\hat{x}$ is an accurate approximation to x.
- For an *a posteriori* error bound, one needs to estimate $\|\hat{r}\|$ and $\kappa(A)$.
- Typically one estimates $\kappa_1(A) = \|A\|_1\|A^{-1}\|_1$ without computing A^{-1}, but allowing LU factorization of A.
  - For any vector $w \in \mathbb{R}^n$ and $\|w\|_1 = 1$, we have the lower bound $\kappa_1(A) \geq \|A\|_1\|A^{-1}w\|_1$.
  - If w has a significant component in the direction near maximum magnification by A^{-1}, then $\kappa_1(A) \approx \|A\|_1\|A^{-1}w\|_1$.
  - Good estimators conduct systematic searches for w that approximately maximizes $\|A^{-1}w\|_1$.